$$P = (T - 1)^2 \quad \text{and} \quad Q = (T + 1)^2$$

P coefficients:
$(1, -2, 1)$

Q coefficients:
$(1, 2, 1)$

Nonzero middle coefficient difference: 4

mod 2	mod 3	mod 5	mod 7
P: 0	P: 1	P: 3	P: 5
Q: 0	Q: 2	Q: 2	Q: 2
collapse	separate	separate	separate